Physics 829: Homework Set No. 5

Due date: Wednesday, May 11, 2012, 1:00pm in PRB M2043 (Biao Huang's office)

Total point value of set: 70 points

Problem 1 (5 pts.):

Show by explicit differentiation that $G(\boldsymbol{r}, \boldsymbol{r'}) = e^{ik|\boldsymbol{r}-\boldsymbol{r'}|}/(4\pi|\boldsymbol{r}-\boldsymbol{r'}|)$ is the Green function of the Helmholtz equation, i.e. it solves the Helmholtz equation for a point source.
Hint: Think carefully about how to separate the δ-function piece from the regular terms.

Problem 2 (5 pts.): Exercise 19.3.1 (Shankar, p. 533)

Problem 3 (10 pts.): Exercise 19.3.2 (Shankar, p. 533)

Problem 4 (10 pts.): Exercise 19.3.3 (Shankar, p. 533)

Problem 5 (10 pts.): Exercise 19.5.1 (Shankar, p. 546)

Problem 6 (5 pts.): Exercise 19.5.2 (Shankar, p. 553)

Problem 7 (25 pts.): Exercise 19.5.4 (Shankar, p. 554)

Problem 1

Use $\Delta\left(\frac{1}{r}\right) = -4\pi\delta(\vec{r})$

$$\Delta\left(\frac{e^{ik|\vec{r}-\vec{r}'|}}{4\pi|\vec{r}-\vec{r}'|}\right) = \frac{1}{4\pi}\Delta\left(\frac{1}{|\vec{r}-\vec{r}'|}\right) + \frac{1}{4\pi}\Delta\frac{e^{ik|\vec{r}-\vec{r}'|}-1}{|\vec{r}-\vec{r}'|}$$

$$= -\delta(\vec{r}-\vec{r}') + \frac{1}{4\pi}\Delta f(\vec{r}-\vec{r}')$$ where $\lim_{|\vec{r}-\vec{r}'|\to 0} f(\vec{r}-\vec{r}')$ is finite, so $\Delta f(\vec{r}-\vec{r}')$ will not contribute a singular term at $\vec{r}=\vec{r}'$

Proceed with $\Delta f(\vec{r}-\vec{r}')$. Introduce $\vec{\rho} = \vec{r}-\vec{r}'$ and use

$\vec{\nabla}_\rho f(\vec{\rho}) = \vec{\nabla}_r f(\vec{r}-\vec{r}')$ (chainrule). Use $\Delta_\rho f = \frac{1}{\rho}\frac{\partial^2}{\partial\rho^2}(\rho f)$ + angular derivatives.

This is safe for $f(\vec{\rho})$, since it is finite at $\vec{\rho}=0$, but not for the original fct. $\frac{e^{ik\rho}}{\rho}$ which is not.

$$\Delta_\rho \frac{e^{ik\rho}-1}{\rho} = \frac{1}{\rho}\frac{\partial^2}{\partial\rho^2}\left(e^{ik\rho}-1\right) = (-k^2)\frac{e^{ik\rho}}{\rho}.$$

$$\Rightarrow \Delta_\rho\left(\frac{e^{ik\rho}}{4\pi\rho}\right) = -\delta(\vec{\rho}) - k^2\frac{e^{ik\rho}}{4\pi\rho}$$

$$\Rightarrow \boxed{\left(\Delta_r + k^2\right)\frac{e^{ik|\vec{r}-\vec{r}'|}}{4\pi|\vec{r}-\vec{r}'|} = -\delta(\vec{r}-\vec{r}')}$$ as desired.

Exercise 19.3.1

$$\left(\frac{d\sigma}{d\Omega}\right)_{Yukawa} = \frac{4\mu^2 g^2}{\hbar^4} \frac{1}{\left[\mu_0^2 + 4k^2\sin^2(\frac{\vartheta}{2})\right]^2} \quad \text{for } V(r) = \frac{g e^{-\mu_0 r}}{r}$$

Write $r_0 = 1/\mu_0$ and $2\sin^2(\frac{\vartheta}{2}) = 1-\cos\vartheta$:

$$V(r) = \frac{g e^{-r/r_0}}{r}$$

$$\frac{d\sigma}{d\Omega} = 4r_0^2 \left(\frac{\mu g r_0}{\hbar^2}\right)^2 \frac{1}{\left[1 + 2k^2 r_0^2 (1-\cos\vartheta)\right]^2}$$

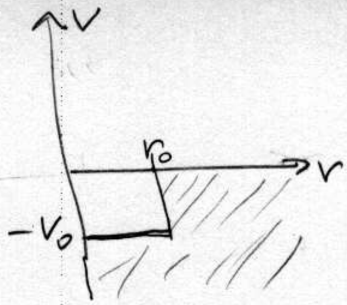
$$\sigma = \int \frac{d\sigma}{d\Omega} d\Omega = 2\pi \int_{-1}^{1} d(\cos\vartheta) \; 4r_0^2 \left(\frac{\mu g r_0}{\hbar^2}\right)^2 \frac{1}{(\underbrace{1+2k^2r_0^2 - 2k^2r_0^2\cos\vartheta}_{u})^2}$$

$$= 8\pi \left(\frac{\mu g r_0}{\hbar^2}\right)^2 \frac{-1}{2k^2r_0^2} \int_{1+4k^2r_0^2}^{1} du \, \frac{1}{u^2}$$

$$= 8\pi \left(\frac{\mu g r_0}{\hbar^2}\right)^2 \frac{1}{2k^2r_0^2} \left(\frac{1}{u}\right)_{1+4k^2r_0^2}^{1} = 8\pi \left(\frac{\mu g r_0}{\hbar^2}\right)^2 \frac{1}{2k^2r_0^2}\left(1 - \frac{1}{1+4k^2r_0^2}\right)$$

$$= 8\pi \left(\frac{\mu g r_0}{\hbar^2}\right)^2 \frac{1}{2k^2r_0^2} \frac{4k^2r_0^2}{1+4k^2r_0^2} = 16\pi \left(\frac{\mu g r_0}{\hbar^2}\right)^2 \frac{1}{1+4k^2r_0^2} \checkmark$$

Exercise 19.3.2



$$V(r) = -V_0\,\Theta(r_0 - r)$$
spherically symmetric

(1) $$\frac{d\sigma}{d\Omega} = |f(\vartheta)|^2 \overset{(19.3.8)}{=} \frac{4\mu^2}{\hbar^2 q^2}\left|\int_0^\infty \sin(qr')\,V(r')\,r'dr'\right|$$

$$= \frac{4\mu^2 V_0^2}{\hbar^2 q^2}\left|\int_0^{r_0} dr'\,r'\,\sin(qr')\right|^2$$

$$= \frac{4\mu^2 V_0^2}{\hbar^2 q^2}\left|\frac{d}{dq}\int_0^{r_0} dr'\,\cos(qr')\right|^2$$

$$= \frac{4\mu^2 V_0^2}{\hbar^2 q^2}\left|\frac{d}{dq}\left(\frac{\sin(qr')}{q}\right)_0^{r_0}\right| = \frac{4\mu^2 V_0^2}{\hbar^2 q^2}\left|\frac{d}{dq}\left(\frac{\sin(qr_0)}{q}\right)\right|^2$$

$$= \frac{4\mu^2 V_0^2}{\hbar^2 q^2}\left(\frac{r_0\cos(qr_0)}{q} - \frac{\sin(qr_0)}{q^2}\right)^2$$

$$= \frac{4\mu^2 V_0^2}{\hbar^2 q^6}\left(\sin(qr_0) - (qr_0)\cos(qr_0)\right)^2$$

$$= 4r_0^2\left(\frac{\mu V_0 r_0^2}{\hbar^2}\right)^2 \frac{\left(\sin(qr_0) - (qr_0)\cos(qr_0)\right)^2}{(qr_0)^6} \quad \checkmark$$

(2) For $kr_0 \to 0$, $qr_0 = 2kr_0 \sin\frac{\vartheta}{2} \to 0$. The angular dependence of $\frac{d\sigma}{d\Omega}$ comes from the factor $\frac{(\sin(qr_0) - (qr_0)\cos(qr_0))^2}{(qr_0)^6}$.

Compute

$$\lim_{x\to 0}\frac{(\sin x - x\cos x)^2}{x^6} = \lim_{x\to 0}\frac{\left((x - \frac{x^3}{3!}) - x(1 - \frac{x^2}{2}) + O(x^5)\right)^2}{x^6} =$$

$$= \lim_{x\to 0} \frac{\left(x^3/3 + O(x^5)\right)^2}{x^6} = \frac{1}{9}$$

$$\Rightarrow \lim_{kr_0\to 0} \frac{d\sigma}{d\Omega} = \frac{4}{9} r_0^2 \left(\frac{\mu V_0 r_0^2}{\hbar^2}\right)^2 \quad \text{isotropic} \checkmark$$

$$\Rightarrow \lim_{kr_0\to 0} \sigma = 4\pi \lim_{kr_0\to 0} \frac{d\sigma}{d\Omega} = \frac{16\pi r_0^2}{9}\left(\frac{\mu V_0 r_0^2}{\hbar^2}\right)^2 \quad \checkmark$$

Exercise 19.3.3

$$V(r) = V_0 e^{-r^2/r_0^2}$$

$$\frac{d\sigma}{d\Omega} = |f(\vartheta)|^2 \overset{(19.3.8)}{=} \frac{4\mu^2 V_0^2}{\hbar^2 q^2}\left| \underbrace{\int_0^\infty e^{-r^2/r_0^2} \underbrace{\frac{e^{iqr}-e^{-iqr}}{2i}}_{\sin(qr)} r\,dr}_{\text{even fct.}} \right|^2$$

$$= \frac{\mu^2 V_0^2}{4\hbar^2 q^2}\left| \int_{-\infty}^{\infty} e^{-r^2/r_0^2}\left(e^{iqr}-e^{-iqr}\right) r\,dr \right|^2$$

$$= \frac{\mu^2 V_0^2 r_0^4}{4\hbar^2 q^2}\left| \int_{-\infty}^{\infty} \xi\,d\xi\, e^{-\xi^2}\left(e^{iqr_0\xi}-e^{-iqr_0\xi}\right) \right|^2$$

$$= \frac{\mu^2 V_0^2 r_0^4}{4\hbar^2 q^2}\, e^{-\frac{q^2 r_0^2}{2}}\left| \int_{-\infty}^{\infty} \xi\,d\xi\left(e^{-(\xi-\frac{iqr_0}{2})^2}-e^{-(\xi+\frac{iqr_0}{2})^2}\right) \right|^2$$

$$= \frac{\mu^2 V_0^2 r_0^4}{4\hbar^2 q^2}\, e^{-q^2 r_0^2/2}\left| \underbrace{\int_{-\infty}^{\infty} e^{-t^2}\left(\overset{0}{\cancel{t}}+\frac{iqr_0}{2}-\left(\overset{0}{\cancel{t}}-\frac{iqr_0}{2}\right)\right)dt}_{\substack{\text{(symmetric integration)} \\ iqr_0\sqrt{\pi}}} \right|^2$$

$$= \frac{\mu^2 V_0^2 r_0^4}{4\hbar^2 q^2}\, e^{-q^2 r_0^2/2}\, q^2 r_0^2 \pi = \frac{\pi r_0^2}{4}\left(\frac{\mu V_0 r_0^2}{\hbar^2}\right)^2 e^{-q^2 r_0^2/2} \quad ✓$$

$$= \frac{\pi r_0^2}{4}\left(\frac{\mu V_0 r_0^2}{\hbar^2}\right)^2 e^{-k^2 r_0^2(1-\cos\vartheta)}$$

$$\sigma = \int \frac{d\sigma}{d\Omega}\,d\Omega = \frac{\pi^2 r_0^2}{2}\left(\frac{\mu V_0 r_0^2}{\hbar^2}\right)\int_{-1}^{1} dz\, e^{-k^2 r_0^2(1-z)}$$

$$= \frac{(\pi r_0)^2}{2}\left(\frac{\mu V_0 r_0^2}{\hbar^2}\right)^2 e^{-k^2 r_0^2}\underbrace{\int_{-1}^{1} dz\, e^{k^2 r_0^2 z}}_{\frac{1}{k^2 r_0^2}\left(e^{k^2 r_0^2}-e^{-k^2 r_0^2}\right)} = \frac{\pi^2}{2k^2}\left(\frac{\mu V_0 r_0^2}{\hbar^2}\right)^2\left(1-e^{-2k^2 r_0^2}\right) \quad ✓$$

Exercise 19.5.1

$T_{neutron} = \frac{p^2}{2m} = 100\ MeV \qquad m_{neutron} c^2 = 940\ MeV$

$r_0 = 10^{-15}\ m = 1\ fm$

$l_{max} = k r_0$

$p = \sqrt{2mT} \qquad \Rightarrow \quad l_{max} = \frac{\sqrt{2mT}}{\hbar} r_0 = \frac{\sqrt{2mc^2 T}}{\hbar c} r_0$

$$= \frac{\sqrt{2 \times (940\ MeV) \times 100\ MeV}}{200\ MeV\ fm} \cdot 1\ fm =$$

$$= \sqrt{\frac{940}{200}} \approx \sqrt{4.7} \approx 2.18$$

$\longrightarrow \boxed{l_{max} \simeq 2}$ ✓

Exercise 19.5.2

$$f(\vartheta) = \frac{1}{k}\sum_{l=0}^{\infty}(2l+1)\, e^{i\delta_l}\sin\delta_l\, P_l(\cos\vartheta) \qquad \text{from Eq. (19.5.17)}$$

$$\sigma = \int |f(\vartheta)|^2 d\Omega = 2\pi\frac{1}{k^2}\sum_{ll'}(2l+1)(2l'+1)\, e^{i(\delta_l-\delta_{l'})}\sin\delta_l\sin\delta_{l'} \underbrace{\int_{-1}^{1} d\cos\vartheta\, P_l(\cos\vartheta)\, P_{l'}(\cos\vartheta)}_{\frac{2}{2l+1}\delta_{ll'}}$$

$$= \frac{4\pi}{k^2}\sum_l (2l+1)\sin^2\delta_l \quad \checkmark$$

$$\mathrm{Im}\, f(\vartheta=0) = \frac{1}{k}\sum_{l=0}^{\infty}(2l+1)\,\mathrm{Im}(\cos\delta_l + i\sin\delta_l)\sin\delta_l\, \underbrace{P_l(1)}_{=1}$$

$$= \frac{1}{k}\sum_{l=0}^{\infty}(2l+1)\sin^2\delta_l$$

$$\Rightarrow \frac{4\pi}{k}\,\mathrm{Im}\, f(0) = \frac{4\pi}{k^2}\sum_{l=0}^{\infty}(2l+1)\sin^2\delta_l = \sigma \quad \checkmark$$

Exercise 19.5.4

Show for $V(r) = -V_0\,\Theta(r_0 - r)$, $\boxed{\delta_0 = -kr_0 + \tan^{-1}\left(\frac{k}{k'}\tan(k'r_0)\right)}$

where $k = \frac{1}{\hbar}\sqrt{2\mu E}$ and $k' = \frac{1}{\hbar}\sqrt{2\mu(E+V_0)}$, and

explore the consequences of this formula as you increase V_0.

For $r<r_0$, the radial wave function ($R_{kl} = \frac{u_{kl}}{r}$) for s-waves ($l=0$) solves

$$\left(-\frac{d^2}{dr^2} - \frac{2\mu V_0}{\hbar^2} - k^2\right)u_{k0}(r) = -\left(\frac{d^2}{dr^2} + k'^2\right)u_{k0} = 0$$

$$\Rightarrow u_{k0}(r) = A'\sin(k'r) \qquad \left(\frac{\cos(k'r)}{r} \text{ diverges at } r=0!\right)$$

For $r>r_0$, the solution is

$$u_{k0}(r) = A\,e^{ikr} + B\,e^{-ikr}.$$

The s-wave phase shift is

$$S_0(k) = e^{2i\delta_0} = -\frac{A}{B} \qquad (\text{Eq. } (19.5.38))$$

We get A and B in terms of A' by requiring continuity at $r=r_0$:

$$A'\sin(k'r_0) = A\,e^{ikr_0} + B\,e^{-ikr_0} \qquad (\underline{1})$$

$$A'k'\cos(k'r_0) = ik\left(A\,e^{ikr_0} - B\,e^{-ikr_0}\right) \qquad (\underline{2})$$

$$\to A'\frac{k'}{ik}\cos(k'r_0) = A\,e^{ikr_0} - B\,e^{-ikr_0} \qquad (\underline{3})$$

$(\underline{1}) + (\underline{3})$:

$$A'\left(\sin(k'r_0) + \frac{k'}{ik}\cos(k'r_0)\right) = 2Ae^{ikr_0} \qquad (\underline{4})$$

$(\underline{1}) - (\underline{3})$

$$A'\left(\sin(k'r_0) - \frac{k'}{ik}\cos(k'r_0)\right) = 2Be^{-ikr_0} \qquad (\underline{5})$$

$(\underline{4}) : (\underline{5})$:

$$\frac{\sin(k'r_0) + \frac{k'}{ik}\cos(k'r_0)}{\sin(k'r_0) - \frac{k'}{ik}\cos(k'r_0)} = \frac{A}{B}e^{2ikr_0} = -e^{2i(\delta_0 + kr_0)}$$

$$= -\frac{e^{i(\delta_0+kr_0)}}{e^{-i(\delta_0+kr_0)}} = -\frac{\cos(\delta_0+kr_0) + i\sin(\delta_0+kr_0)}{\cos(\delta_0+kr_0) - i\sin(\delta_0+kr_0)}$$

$$= -\frac{1+i\tan(\delta_0+kr_0)}{1-i\tan(\delta_0+kr_0)}$$

$$\Rightarrow \frac{\frac{ik}{k'}\tan(k'r_0)+1}{\frac{ik}{k'}\tan(k'r_0)-1} = \frac{i\tan(\delta_0+kr_0)+1}{i\tan(\delta_0+kr_0)-1}$$

$$\Rightarrow \boxed{\delta_0 + kr_0 = \tan^{-1}\left(\frac{k}{k'}\tan(k'r_0)\right)}$$ as desired.

We are told to consider small k, $kr_0 \to 0$:

$$\delta_0(k) \xrightarrow[kr_0 \to 0]{} \tan^{-1}\left(\frac{k}{k'}\tan(k'r_0)\right)$$

Whenever δ_0 becomes an odd multiple of $\pi/2$ we hit a resonance.

This happens when $k'r_0$ is an odd multiple of $\frac{\pi}{2}$,

because

$$\delta_0\left(k'r_0=(2n+1)\frac{\pi}{2}\right)=-kr_0+\tan^{-1}\left(\frac{k}{k'}\cdot\infty\right)=-kr_0+(2m+1)\frac{\pi}{2}$$
$$\approx(2m+1)\frac{\pi}{2}$$

So we have resonances at $\boxed{k_n'=(2n+1)\frac{\pi}{2r_0}}$ as claimed.

Near these resonances

$$\frac{k}{k'}\tan(k'r_0)=\frac{k}{k'}\tan\left((2n+1)\frac{\pi}{2}+(k'-k_n')r_0\right)$$
$$=\frac{k}{k'}\frac{\sin\left((2n+1)\frac{\pi}{2}+(k'-k_n)r_0\right)}{\cos(\qquad)}$$
$$=\frac{k}{k'}\frac{\cos((k'-k_n')r_0)}{+\sin((k_n'-k')r_0)}\approx\frac{k}{k'}\frac{1}{(k_n'-k')r_0}$$
$$=\frac{k}{r_0}\cdot\frac{1}{(k_n'k'-k'^2)}\underset{k'\approx k_n'}{\approx}\frac{\hbar^2k/2\mu}{r_0\left(\underbrace{\frac{\hbar^2k_n'^2}{2\mu}}_{E_n+V_0}-\underbrace{\frac{\hbar^2k'^2}{2\mu}}_{E+V_0}\right)}$$
$$\approx\frac{\hbar^2k_n}{2\mu r_0}\frac{1}{E_n-E}=\frac{\Gamma_n/2}{E_n-E}$$

where $\Gamma_n=\frac{\hbar^2k_n}{\mu r_0}$

and $E_n=\frac{\hbar^2k_n^2}{2\mu}$ ✓

$E=\frac{\hbar^2k^2}{2\mu}$

Hence for $k'\simeq k_n'\equiv(2n+1)\frac{\pi}{2r_0}$,

$$\delta_0(k)\simeq\tan^{-1}\left(\frac{\Gamma_n/2}{E_n-E}\right)\quad\text{with}\quad E_n=\frac{\hbar^2k_n^2}{2\mu},\ E=\frac{\hbar^2k^2}{2\mu},\ \Gamma_n=\frac{\hbar^2k_n}{\mu r_0}$$

(I used $\frac{\hbar^2}{2\mu}(k_n'^2-k'^2)=\frac{\hbar^2}{2\mu}(k_n^2-k^2)$ since V_0 drops out from the difference.)

If we start with V_0 too small to support a bound state, then the first resonance occurs at zero energy ($k=0$) when

$$k_1' = \frac{\pi}{2r_0} \quad (n=0) \quad \Longrightarrow \quad \frac{\hbar^2 k_1'^2}{2\mu} = \frac{\hbar^2}{2\mu}\frac{\pi^2}{4r_0^2} = \frac{\hbar^2\pi^2}{8\mu r_0^2}$$

Since $\dfrac{\hbar^2 k_1'^2}{2\mu} = E_1 + V_0 = \dfrac{\hbar^2 k^2}{2\mu} + V_0$, for $k=0$ this

resonance occurs when

$$V_0 = \frac{\hbar^2\pi^2}{8\mu r_0^2}$$

According to exercise 12.6.9 (p. 349), this potential is just deep enough to support a bound state, with zero binding energy.

As we increase V_0, $k=0$ (zero energy) scattering becomes again non-resonant. The next resonance at $k=0$ appears when V_0 becomes large enough that a second bound state appears with zero binding energy.